

Luis Andrés Alemán Sánchez

📍 León, GTO, México ✉ andresaleman142@gmail.com 🔗 andresalemn.github.io
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Summary

Mechatronics Engineer with professional experience developing embedded and robotics software for humanoid and mobile robotic systems. Specialized in C/C++, ROS 2, real-time control, CANopen communication, sensor integration, and system debugging. Experienced in firmware development on Teensy and ESP32 microcontrollers, inverse kinematics, industrial automation, PLC programming, and electrical integration. Strong hands-on background spanning low-level control, hardware bring-up, robotic prototyping, and software-hardware integration. Fluent in Spanish and professionally proficient in English (C1)

Experience

Junior Robotics & Control Engineer

León, Guanajuato, Mexico

Inbiodroid

May 2024 – present

- Developed torque-based joint position controllers in C++ for multiple subsystems of a 20-DOF humanoid robot, achieving sub-millimeter (below 1 mm) positioning accuracy using encoder feedback and custom control algorithms.
- Implemented embedded firmware on Teensy 4.1 microcontrollers and integrated distributed motion-control nodes over the CANopen protocol.
- Designed and tested real-time teleoperation functionality by integrating IMU sensors, inverse kinematics, and UDP-based communication between robotic subsystems.
- Performed system integration, debugging, calibration, and validation of robotic hardware and software components during prototype development.
- Developed PLC control logic with Siemens LOGO!, created electrical schematics in SolidWorks Electrical, and designed pneumatic circuits in FluidSIM for industrial automation projects.

Projects

3-DOF DC Motor Manipulator

University Project

- Designed and implemented embedded motor-control firmware in C++ for a 3-DOF robotic manipulator.
- Integrated quadrature encoders and tuned PID controllers to achieve ± 1 mm positioning accuracy.
- Developed a serial communication interface between the embedded controller and a Python application for monitoring and control.

Alexa-Controlled Robot (ROS 2 + Flask)

Personal Project

- Developed a ROS 2-based robotic control system integrating Amazon Alexa voice commands through a Flask web service and secure tunneling.
- Implemented ROS 2 nodes, lifecycle management, and ros2_control interfaces for real-time servo actuation and state feedback.
- Designed APIs and communication pipelines between cloud services and physical robotic hardware.

Blind Spot Detection System

Bachelor's Capstone Project

- Developed a Raspberry Pi driver-assistance prototype using Python, OpenCV, and ultrasonic sensing.
- Implemented signal-processing and digital-filtering techniques to improve obstacle-detection robustness and reduce sensor noise.
- Integrated computer-vision and distance-sensing modules into a single embedded application.

Skills

Programming: C, C++, Python, MATLAB

Robotics Software: ROS 2, ros2_control, RViz, TF2, URDF, MoveIt, Gazebo, CoppeliaSim, Peter Corke Robotics Toolbox

Embedded Systems: Teensy, ESP32, Arduino, Raspberry Pi, Firmware Development, Real-Time Systems, CANopen, UART, SPI, I²C, UDP, Serial Communication

Control & Modeling: PID Control, Kinematics, Dynamics, Inverse Kinematics, System Modeling, MATLAB/Simulink

Industrial Robotics: ABB, FANUC, Universal Robots, Mitsubishi

Industrial Automation: Siemens LOGO!, CODESYS, SolidWorks Electrical, FluidSIM, Electrical Schematics, Pneumatic Systems

Tools: Git, GitHub, Docker, Linux (Ubuntu), VS Code, PlatformIO, LaTeX

Languages: Spanish (Native), English (C1), Finnish (A2)

Education

Universidad Anáhuac Mayab
BS in Mechatronics Engineering

Mérida, Yucatán, Mexico
Sept 2019 – Dec 2023

Universidad Anáhuac Mayab
Diploma in Cultural Management

Mérida, Yucatán, Mexico
Jan 2020 – May 2023

Completed undergraduate coursework in the Culmen Leadership and Excellence in Arts & Culture Program.

Certificates

Industrial Robotics Manufacturing Systems Integrator

July 2026

- Programming and integration of ABB, FANUC, Universal Robots, and Mitsubishi robotic systems.
- Commissioning, troubleshooting, and validation of robotic cells.
- Application of industrial safety procedures and standards.

Complementary Education

- **Professional Development:** Specialized courses in ROS 2, MoveIt, Modern Robotics, Collaborative Robotics in Industry, Operating Systems, and Computer Networks (Coursera, Udemy, edX). Motion Planning — Invited participant in a graduate-level course at CIMAT (2026).
- **International Experience:** Selected to participate in academic exchange programs in Finland (2017–2018) and Canada (2022).